Downstream development of extratropical transition of tropical cyclones in the Southern Hemisphere

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1. Introduction and motivation

- Tropical cyclones (TCs) that move into the midlatitudes undergo structural changes and transform into extratropical cyclones, known as extratropical transition (ET). ET can produce severe weather locally and can also influence downstream weather. During this process, the transitioning cyclone directly interacts with the midlatitude flow, and its impacts can propagate downstream by modifying upper-level Rossby wave packets (Keller et al. 2019). Although the importance of downstream impacts has been well recognized in the Northern Hemisphere, relatively few studies have examined this issue in the Southern Hemisphere.

2. Data and method

As a follow-up work of Jin et al. (2026), we use an objective cyclone tracking and classification framework (Han and Ullrich, 2025) on ERA5 reanalysis to identify ET events over the Southern Hemisphere. To address the case-to-case variability of synoptic configuration at ET, K-means clustering is applied to MSLP and Z500 fields, which identifies four unique ET clusters (Fig. 1). The downstream development of individual cluster is studied in the current work.

3. Results

3.1. ET clusters

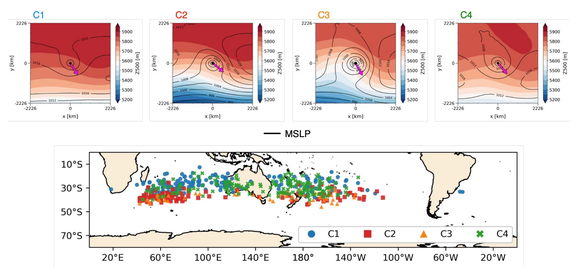


Fig. 1 MSLP and Z500 for the four ET clusters (top) and their geographic locations (bottom).

3.2. Downstream development for ET clusters

C2 & C3
- Pronounced downstream anticyclonic PV anomalies and blocking frequency (Fig. 2b and c)

C1 & C4
- Weak anticyclonic PV anomalies (Fig. 2a and d)

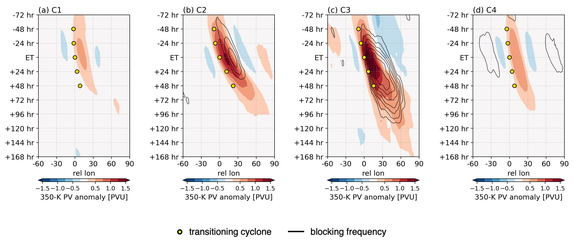


Fig. 2 Composite Hovmöller diagrams of 350-K PV anomaly and blocking frequency centred at ET time and position for the four clusters (a–d).

3.3. Potential vorticity (PV) budget analysis

A PV budget analysis is applied to understand the dynamics of downstream development associated with C2 and C3 ET events. Following Barnes et al. (2025), the PV budget is constructed as:

$$\partial_\theta q'/\partial t + \overline{\boldsymbol{V}_\phi} \cdot \nabla_\theta q' + \boldsymbol{V}'_\phi(\partial_\theta \bar{q}/\partial y) = -\boldsymbol{V}_\chi \cdot \nabla_\theta q - \boldsymbol{V}'_\phi \cdot \nabla_\theta q' = I_\chi + I'_\phi \qquad (1)$$

Eq. (1) describes the barotropic Rossby wave on isentropic levels, forced by I_χ and I'_ϕ. The I_χ measures the interaction between diabatic heating and waveguide, and the I'_ϕ measures the nonlinearities, and $[\overline{\boldsymbol{V}_\phi} \cdot \nabla_\theta q' + \boldsymbol{V}'_\phi(\partial_\theta \bar{q}/\partial y)]$ describes the propagation of Rossby wave.

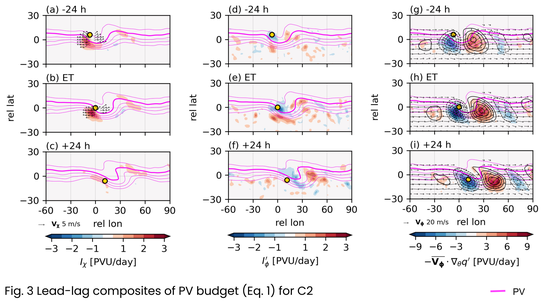


Fig. 3 Lead-lag composites of PV budget (Eq. 1) for C2

C2
- Immediate downstream ridge development (Fig. 3 a-c)
- Strong nonlinear interaction between TC and upstream trough (Fig. 3 d-f)
- Eastward-propagating wave packets (Fig. 3 g-i)

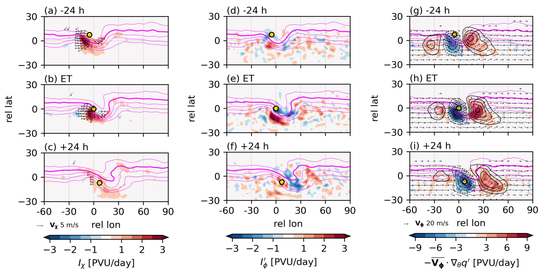


Fig. 4 As in Fig.3 but for C3.

C3
- Divergent outflow anchors the eastward propagating wave packets (Fig. 4 a-c)
- Strong I_χ and I'_ϕ values coincide highly amplified downstream ridge (Fig. 4 d-f)
- Wave packets stop propagating at 60 deg east (Fig. 4g-i)

3.4. Eddy kinetic energy (*Ke*) budget analysis

Downstream development of ET can also be explained from an energetics perspective. From Orlanski and Sheldon (1995), the *Ke* budget can be expressed as:

$$\frac{1}{g}\int_{1000}^{100}\frac{\partial K_e}{\partial t}\,dp = \frac{1}{g}\int_{1000}^{100}\left[-\nabla_p\cdot(\vec{v}\phi)_a - \omega\alpha - \nabla_p\cdot(\overrightarrow{V K_e}) - \vec{v}\cdot(\vec{v}\cdot\nabla_p\overrightarrow{V_m}) + residue\right]dp \qquad (2)$$

Eq. (2) describes the change in Ke is associated with divergence of ageostrophic eddy flux, baroclinic conversion, divergence of Ke flux, and barotropic conversion.

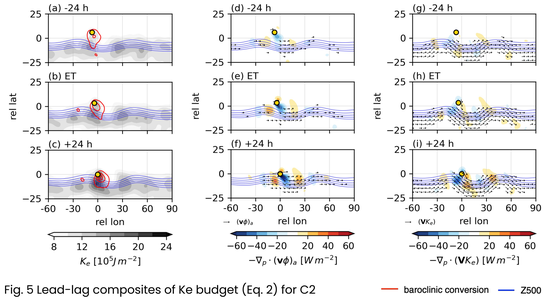


Fig. 5 Lead-lag composites of Ke budget (Eq. 2) for C2

C2
- Ke accumulates rapidly downstream (Fig. 5 a-c)
- TC brings additional energy to the midlatitude flow (Fig. 5 a-c)
- The additional energy is redistributed by ageostrophic flow, which propagates further downstream (Fig. 5 d-i)

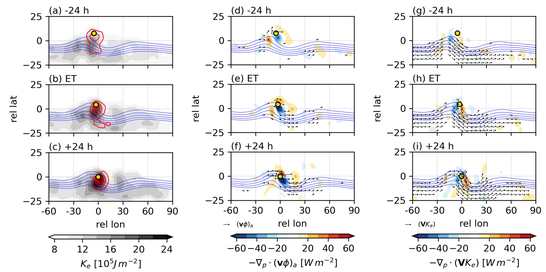
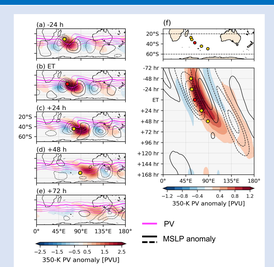


Fig. 6 As in Fig.5 but for C3.

C3
- TC brings more energy into the midlatitudes (Fig. 6 a-c)
- However, the energy propagation is limited in a smaller region downstream of the transitioning TC (Fig. 6 d-i)

4. Conclusion

- C2 and C3 ET events feature different pathways of downstream development (remote versus localised)
- From a **PV viewpoint**: the diabatic processes and interaction between TC and midlatitude flow features forces the Rossby wave propagating downstream (Figs. 3 and 4)
- From a **Ke viewpoint**: TC brings additional energy to the midlatitude flow, redistributed by ageostrophic flow and propagating downstream (Figs. 4 and 5)
- ET events in the southwest Indian Ocean can affect weather downstream in the Australian region, cause heavy rainfall days in the southeast of Australia and warm days in the east of the country after ET (Fig. 7)



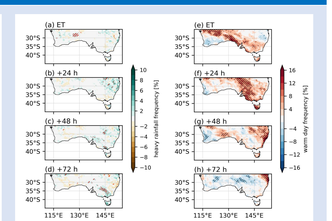


Fig. 7 Downstream influence of C2 and C3 ET events in the southwest Indian Ocean

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